```
> restart;#The Flux of torsion particle Pto
> de := 1/V^2 ·H(r)·diff(u(t),t,t) + u(t)·diff(H(r),r,r) = 1/H4(z);
```

$$de := \frac{H(r)\left(\frac{d^2}{dt^2}\,u(t)\right)}{V^2} + u(t)\left(\frac{d^2}{dr^2}\,H(r)\right) = \frac{1}{H4(z)} \tag{1}$$

```
> dsolve(de, u(t));
```

$$u(t) = \frac{e^{_C2\left(\frac{d^2}{dr^2}\,H(r)\right)H4(z)}\cos\left(\frac{\sqrt{H(r)\left(\frac{d^2}{dr^2}\,H(r)\right)}\,V\,(_C1+t)}{H(r)}\right)+1}{\left(\frac{d^2}{dr^2}\,H(r)\right)H4(z)} \tag{2}$$

```
> eq1 := r·diff(H4(r),r,r) + diff(H4(r),r) + (ν^2·r - b^2·r^2/R)·H4(r) = 0;
```

$$eq1 := r\left(\frac{d^2}{dr^2}\,H4(r)\right) + \frac{d}{dr}\,H4(r) + \left(\nu^2\,r - \frac{b^2\,r^2}{R}\right)H4(r) = 0 \tag{3}$$

```
> eq2 := (r·diff(H4(r),r,r) + diff(H4(r),r) + (ν^2·r - b^2·r^2/R)·H4(r))/r = 0;
```

$$eq2 := \frac{r\left(\frac{d^2}{dr^2}\,H4(r)\right) + \frac{d}{dr}\,H4(r) + \left(\nu^2\,r - \frac{b^2\,r^2}{R}\right)H4(r)}{r} = 0 \tag{4}$$

```
>   # Thus with R:=3.566·10^22 m  we can  ignore the therm - b^2·r/R for 1/R < r < R;
```

$$\frac{1}{R} < r < R \tag{5}$$

```
> 1/r diff((r·diff H(r),r),r) + k^2· H(r) = 0;
```

$$\frac{2\,diff\left(\frac{d}{dr}\,H(r)\right) + r\,diff\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + k^2\,H(r) = 0 \tag{6}$$

```
> de := 1/r·diff(r·diff(H(r),r),r) + k^2·H(r);
```

$$de := \frac{\frac{d}{dr}\,H(r) + r\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + k^2\,H(r) \tag{7}$$

```
> bc := H(2 R) = 0, D(H)(1/R) = R;
```

$$bc := H(2\,R) = 0,\ D(H)\left(\frac{1}{R}\right) = R \tag{8}$$

```
> dsol := dsolve( {de, bc}, H(r) );
```

$$dsol := H(r) = \frac{R\,\mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} - \frac{\mathrm{BesselJ}(0, 2\,k\,R)\,R\,\mathrm{BesselY}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \tag{9}$$

```
> H(r) := R*BesselY(0, 2*k*R)*BesselJ(0, k*r)/(k*(BesselJ(0, 2*k*R)*BesselY(1, k/R) - BesselY(0, 2*k*R)*BesselJ(1, k/R))) - BesselJ(0, 2*k*R)*R*BesselY(0, k*r)/(k*(BesselJ(0, 2*k*R)*BesselY(1, k/R) - BesselY(0, 2*k*R)*BesselJ(1, k/R)));
```

$$H := r \mapsto \frac{R\cdot\mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}(0, k\cdot r)}{k\cdot\left(\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} - \frac{\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot R\cdot\mathrm{BesselY}(0, k\cdot r)}{k\cdot\left(\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \tag{10}$$

```
> u(t) := (exp(_C2*(diff(H(r), r$2))*H4(z))*cos(sqrt((diff(H(r), r$2))*H(r))*V(_C1 + t)/H(r)) + 1)/((diff(H(r), r$2))*H4(z));
```

$$u := t \mapsto \frac{e^{_C2\cdot\left(\frac{d^2}{dr^2}\,H(r)\right)\cdot H4(z)}\cdot\cos\left(\frac{\sqrt{\left(\frac{d^2}{dr^2}\,H(r)\right)\cdot H(r)}\cdot V\cdot(_C1 + t)}{H(r)}\right) + 1}{\left(\frac{d^2}{dr^2}\,H(r)\right)\cdot H4(z)} \tag{11}$$

```
> B := subs(H(r) = R*BesselY(0, k*R)*BesselJ(0, k*r)/(k*(BesselJ(0, k*R)*BesselY(1, k/R) - BesselY(0, k*R)*BesselJ(1, k/R))), H4(z) = sin(z/(2*Rs)), u(t)/r);
```

$$B := \tag{12}$$

$$\left(e^{_C2 \left(- \frac{R\,\mathrm{BesselY}(0, 2kR) \left(\mathrm{BesselJ}(0, kr) - \frac{\mathrm{BesselJ}(1, kr)}{kr} \right) k}{\mathrm{BesselJ}(0, 2kR)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2kR)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)} \right.} \right.$$

$$\left. + \frac{\mathrm{BesselJ}(0, 2kR)\,R \left(\mathrm{BesselY}(0, kr) - \frac{\mathrm{BesselY}(1, kr)}{kr} \right) k}{\mathrm{BesselJ}(0, 2kR)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2kR)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)} \right)$$

$$\sin\left(\frac{z}{2Rs}\right) \cos\left(1 \Big/ \left(R\,\mathrm{BesselY}(0, kR)\,\mathrm{BesselJ}(0, \right. \right.$$

$$\left. \left. kr \right) \right)$$

$$\left(\left(\left(\left(- \frac{R\,\mathrm{BesselY}(0, 2kR) \left(\mathrm{BesselJ}(0, kr) - \frac{\mathrm{BesselJ}(1, kr)}{kr} \right) k}{\mathrm{BesselJ}(0, 2kR)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2kR)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)} \right. \right. \right. \right.$$

$$\left. \left. \left. \left. + \frac{\mathrm{BesselJ}(0, 2kR)\,R \left(\mathrm{BesselY}(0, kr) - \frac{\mathrm{BesselY}(1, kr)}{kr} \right) k}{\mathrm{BesselJ}(0, 2kR)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2kR)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)} \right) R\,\mathrm{BesselY}(0, \right. \right. \right. \right.$$

$$k\,R)\ \mathrm{BesselJ}(0,k\,r)\Bigg)\Bigg/\Bigg(k\left(\mathrm{BesselJ}(0,k\,R)\ \mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,$$

$$k\,R)\ \mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)\Bigg)^{1/2}\Bigg)\ V(_C1+t)\,k\left(\mathrm{BesselJ}(0,k\,R)\ \mathrm{BesselY}\left(1,\frac{k}{R}\right)\right.$$

$$\left.-\mathrm{BesselY}(0,k\,R)\ \mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)\Bigg)+1\Bigg)\Bigg/\Bigg(\left(\left(\right.\right.$$

$$-\frac{R\,\mathrm{BesselY}(0,2\,k\,R)\left(\mathrm{BesselJ}(0,k\,r)-\dfrac{\mathrm{BesselJ}(1,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,2\,k\,R)\ \mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\ \mathrm{BesselJ}\left(1,\frac{k}{R}\right)}$$

$$\left.+\frac{\mathrm{BesselJ}(0,2\,k\,R)\ R\left(\mathrm{BesselY}(0,k\,r)-\dfrac{\mathrm{BesselY}(1,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,2\,k\,R)\ \mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\ \mathrm{BesselJ}\left(1,\frac{k}{R}\right)}\right)$$

$$\left.\sin\left(\frac{z}{2\,Rs}\right)r\right)$$

=

> $$\frac{B\cdot H(r)\cdot\sin\left(\dfrac{z}{2\cdot Rs}\right)}{4\cdot\mathrm{Pi}\cdot r^2};$$

$$\left(\left(\left(

$$\left| k\,R)\ \mathrm{BesselJ}(0,\,k\,r)\right| \Big/ \left(k\left(\mathrm{BesselJ}(0,\,k\,R)\ \mathrm{BesselY}\left(1,\,\frac{k}{R}\right)-\mathrm{BesselY}(0,\right.\right.$$

$$\left.\left.k\,R)\ \mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)\right)\right)^{1/2}\left|\ V\,(_C1+t)\,k\left(\mathrm{BesselJ}(0,\,k\,R)\ \mathrm{BesselY}\left(1,\,\frac{k}{R}\right)\right.\right.$$

$$\left.\left.-\ \mathrm{BesselY}(0,\,k\,R)\ \mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)\right)\right|+1\Bigg|$$

$$\left(\frac{R\ \mathrm{BesselY}(0,\,2\,k\,R)\ \mathrm{BesselJ}(0,\,k\,r)}{k\left(\mathrm{BesselJ}(0,\,2\,k\,R)\ \mathrm{BesselY}\left(1,\,\frac{k}{R}\right)-\mathrm{BesselY}(0,\,2\,k\,R)\ \mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)\right)}\right.$$

$$\left.-\frac{\mathrm{BesselJ}(0,\,2\,k\,R)\ R\ \mathrm{BesselY}(0,\,k\,r)}{k\left(\mathrm{BesselJ}(0,\,2\,k\,R)\ \mathrm{BesselY}\left(1,\,\frac{k}{R}\right)-\mathrm{BesselY}(0,\,2\,k\,R)\ \mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)\right)}\right)\Bigg/$$

$$\left(4\left(-\frac{R\ \mathrm{BesselY}(0,\,2\,k\,R)\left(\mathrm{BesselJ}(0,\,k\,r)-\frac{\mathrm{BesselJ}(1,\,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,\,2\,k\,R)\ \mathrm{BesselY}\left(1,\,\frac{k}{R}\right)-\mathrm{BesselY}(0,\,2\,k\,R)\ \mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)}\right.\right.$$

$$\left.\left.+\frac{\mathrm{BesselJ}(0,\,2\,k\,R)\ R\left(\mathrm{BesselY}(0,\,k\,r)-\frac{\mathrm{BesselY}(1,\,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,\,2\,k\,R)\ \mathrm{BesselY}\left(1,\,\frac{k}{R}\right)-\mathrm{BesselY}(0,\,2\,k\,R)\ \mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)}\right)r^3\,\pi\right)$$